

Ch 6: Supply, Demand, and Government Policies

Price ceilings, price floors, and who really pays a tax

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EC 111 Principles of Microeconomics, Bentley University, Fall 2026

From markets to policy

Chapters 4-5

Markets set P and Q on their own

Elasticity: how far each one moves

Chapter 6

The government steps in:

- a price ceiling

- a price floor

- a tax

Massachusetts minimum wage:

\$15.00 an hour

Federal: \$7.25, unchanged since 2009

Rent control: banned statewide by a 1994 ballot question, back in the debate

Question

Who gains, who loses, and does the policy do what it promises?

Monday: price controls · Wednesday: taxes and who bears them

Two kinds of price control

Price ceiling

Legal **maximum** price

Rent control, gasoline 1973–79,
price-gouging laws

Price floor

Legal **minimum** price

Minimum wage, farm price
supports

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Binding or not?

Ceiling **above** the equilibrium price, floor **below** it: not binding, nothing changes

Ceiling **below** equilibrium, floor **above** it: binding, the market cannot clear

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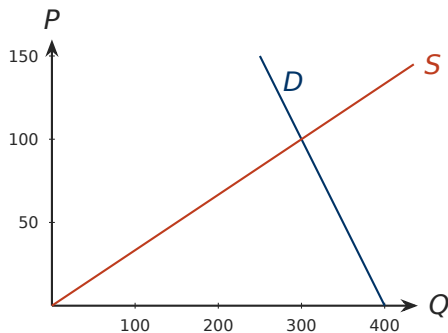
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Who asks for which: buyers lobby for ceilings, sellers for floors.

The market we will use

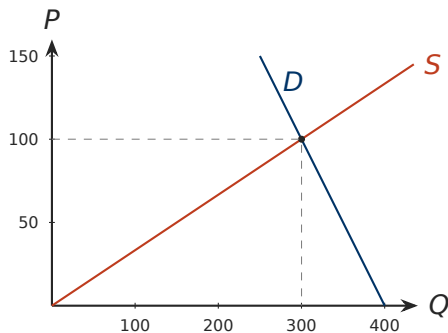


$$Q_d = 400 - P \quad Q_s = 3P$$

Question

Find the equilibrium price and quantity.

The market we will use



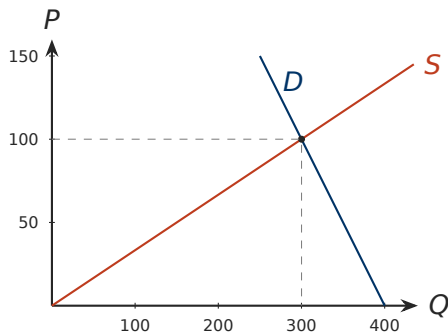
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Find the equilibrium price and quantity.

$$400 - P = 3P \Rightarrow P^* = 100, \quad Q^* = 300$$

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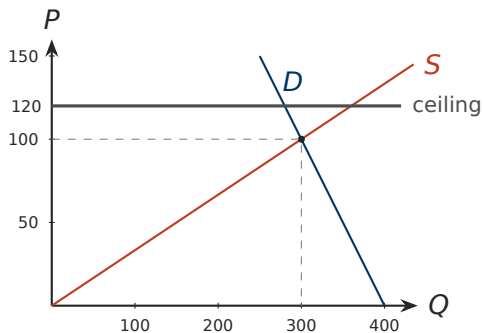
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Every policy today starts from here.

A ceiling at \$120: nothing happens

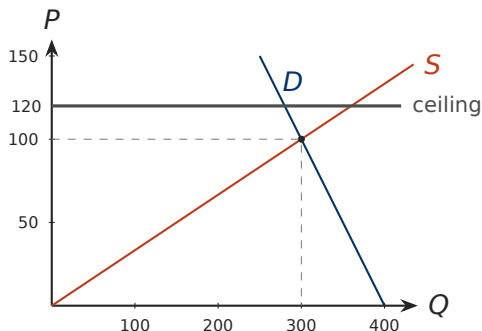


Legal maximum \$120

Market price \$100: already legal

$P = 100$, $Q = 300$, as before

A ceiling at \$120: nothing happens



Legal maximum \$120

Market price \$100: already legal

$P = 100$, $Q = 300$, as before

Not binding

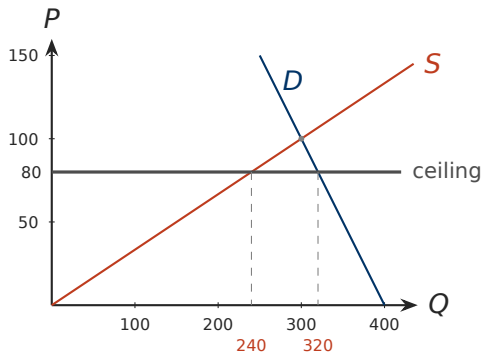
A ceiling above the equilibrium price has **no effect**

A ceiling at \$80: a shortage

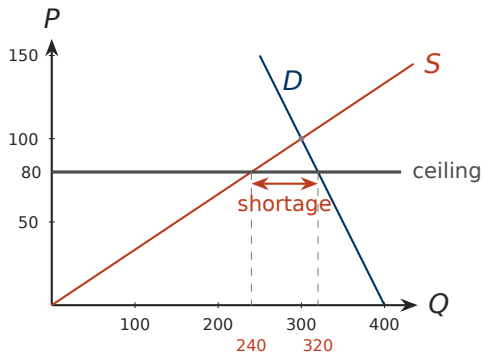
Ceiling \$80, below $P^* = 100$

$$Q_s = 3 \times 80 = 240$$

$$Q_d = 400 - 80 = 320$$



A ceiling at \$80: a shortage



Ceiling \$80, below $P^* = 100$

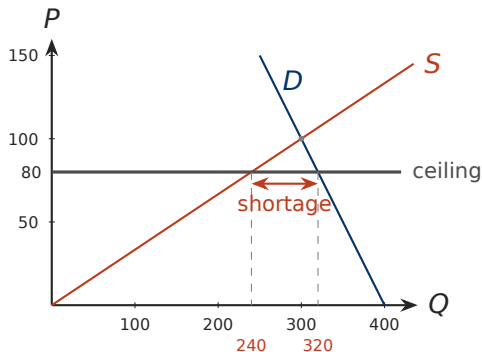
$$Q_s = 3 \times 80 = 240$$

$$Q_d = 400 - 80 = 320$$

$$\text{Shortage} = 320 - 240 = 80$$

Units sold: 240, not 300

A ceiling at \$80: a shortage



Ceiling \$80, below $P^* = 100$

$$Q_s = 3 \times 80 = 240$$

$$Q_d = 400 - 80 = 320$$

$$\text{Shortage} = 320 - 240 = 80$$

Units sold: 240, not 300

Binding

The price cannot rise to clear the market. The shortage **stays**.

If not the price, then what decides who gets it?

320 buyers want it, 240 units exist

Something else does the rationing:

Lines — time wasted, 1973 and 1979 gas stations

Seller's choice — friends, relatives, “no pets, no kids”

Side payments — key money, black markets

Quality falls — the landlord stops fixing things

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A binding ceiling was meant to help buyers. Does it help all of them?

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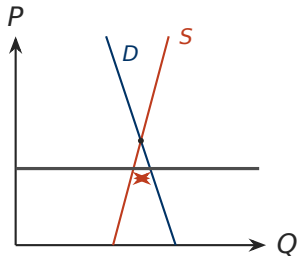
A binding ceiling was meant to help buyers. Does it help all of them?

Buyers who get a unit pay less
80 buyers who wanted one at \$80 get **nothing**

60 units that would have been traded at \$100 are **never produced**

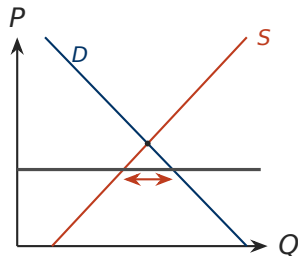
Rent control: short run vs. long run

Short run



steep curves: small shortage

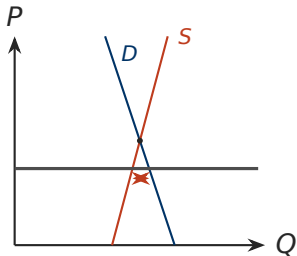
Long run



flat curves: large shortage

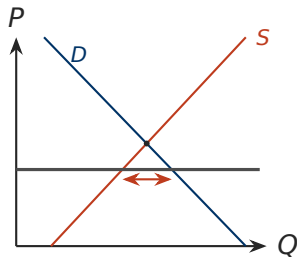
Rent control: short run vs. long run

Short run



steep curves: small shortage

Long run



flat curves: large shortage

Same ceiling. Over years landlords stop building and maintaining; more people want their own place. **Both curves get more elastic**, and the shortage grows.

Rent control: the evidence

Cambridge, MA

Rent control ended 1995 (1994 ballot)

Decontrolled buildings rose in value — and so did **never-controlled** buildings nearby

Autor, Palmer and Pathak, *JPE* 2014

San Francisco

1994: rent control extended to small older buildings

Covered tenants **stayed** longer

Landlords converted units to condos: rental supply in those buildings down about **15%**, rents up city-wide

Diamond, McQuade and Qian, *AER* 2019

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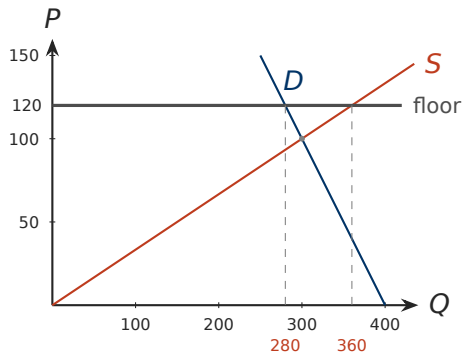
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Today's tenants gain; tomorrow's renters pay, through **fewer units** and **higher rents** elsewhere.

A price floor at \$120: a surplus

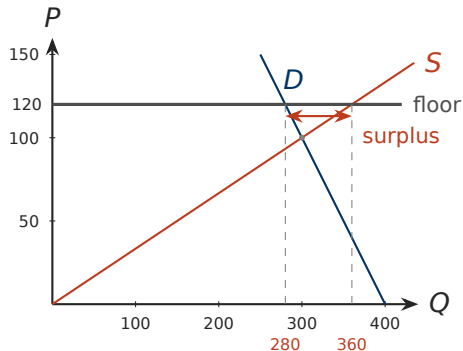


Floor \$120, above $P^* = 100$

$$Q_d = 400 - 120 = 280$$

$$Q_s = 3 \times 120 = 360$$

A price floor at \$120: a surplus



Floor \$120, above $P^* = 100$

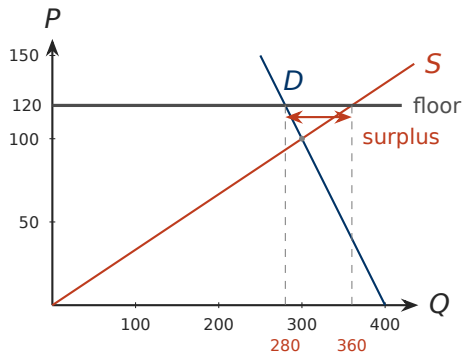
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$$Q_s = 3 \times 120 = 360$$

$$\text{Surplus} = 360 - 280 = 80$$

Units sold: 280, not 300

A price floor at \$120: a surplus



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$$Q_d = 400 - 120 = 280$$

$$Q_s = 3 \times 120 = 360$$

$$\text{Surplus} = 360 - 280 = 80$$

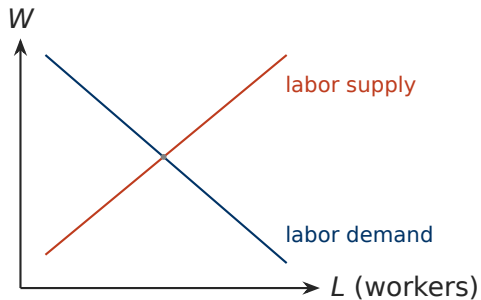
Units sold: 280, not 300

A floor at \$80 would be **not binding**:
\$100 is already above it.

The minimum wage

Price of labor = the **wage**

Firms demand labor, workers supply it

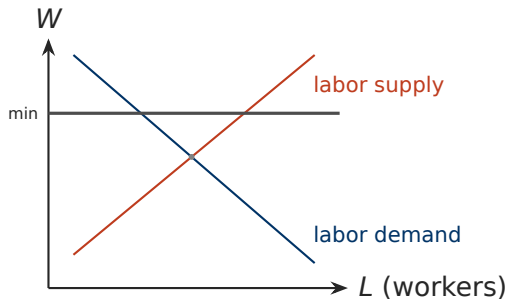


The minimum wage

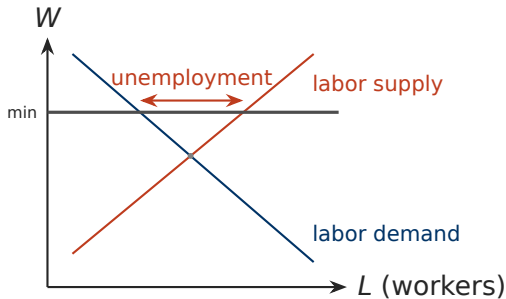
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Minimum above the market wage: a **binding floor**



The minimum wage



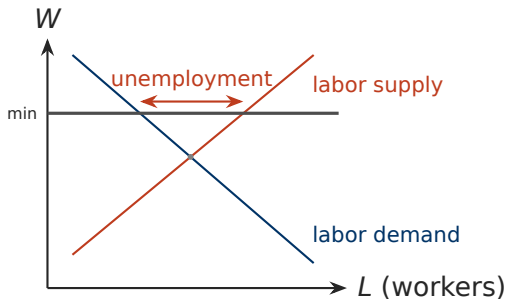
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More want jobs, firms want fewer: **unemployment**

The minimum wage



Price of labor = the **wage**

Firms demand labor, workers supply it

Minimum above the market wage: a **binding floor**

More want jobs, firms want fewer:
unemployment

Binds for young and low-skill workers
Jobs lost depend on the **elasticity of labor demand**

The minimum wage: what the numbers say

Two sides of the same floor

Workers who keep the job: **higher pay**

Workers who lose it or never get hired: **nothing**

Evidence

Card and Krueger (1994): NJ raised it, PA did not — no fall in fast-food jobs

Later work: small effects for modest rises

CBO (2019): a \$15 federal minimum by 2025

Pay raised for **17 million**

1.3 million out of poverty

Jobs lost: median estimate **1.3 million**, range 0 to 3.7 million

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Question

Who wants it raised: a worker who has the job, or a 17-year-old looking for a first one?

Evaluating price controls

The goal is usually fair

Cheaper housing for the poor

A living wage

The tool is blunt

Prices ration and signal (Ch 4,
Hayek)

Fix the price and you lose both

Shortage or surplus, and **fewer
trades**

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Alternatives economists prefer

Rent control → **housing vouchers**:
tenants pay less, landlords still get
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Minimum wage → **Earned Income Tax Credit**: low earners get more,
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The cost moves to taxpayers, and
the market still clears.

Question

Same market: $Q_d = 400 - P$, $Q_s = 3P$.

- (a) A price ceiling at \$60. Binding? Shortage or surplus, and how big? How many units are sold?
- (b) A price floor at \$110. Same questions.
- (c) A price ceiling at \$150. Same questions.

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(c) A price ceiling at \$150. Same questions.

(a) Binding. $Q_s = 180$, $Q_d = 340$: shortage 160, 180 sold.

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(a) Binding. $Q_s = 180$, $Q_d = 340$: shortage 160, 180 sold.

(b) Binding. $Q_d = 290$, $Q_s = 330$: surplus 40, 290 sold.

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(a) Binding. $Q_s = 180$, $Q_d = 340$: shortage 160, 180 sold.

(b) Binding. $Q_d = 290$, $Q_s = 330$: surplus 40, 290 sold.

(c) Not binding: \$100 is legal. $P = 100$, $Q = 300$.

Governments tax goods to raise revenue and to discourage use
Gasoline, cigarettes, alcohol, hotel rooms, Red Sox tickets

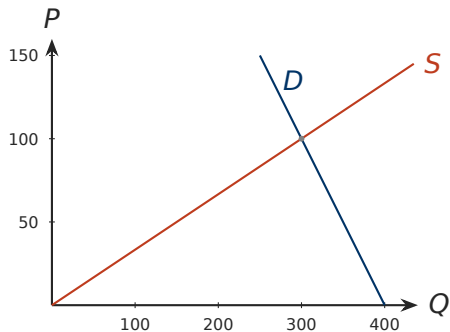
The law says who **sends** the check
The market decides who **pays**

Tax incidence

How the burden of a tax is **shared** between buyers and sellers

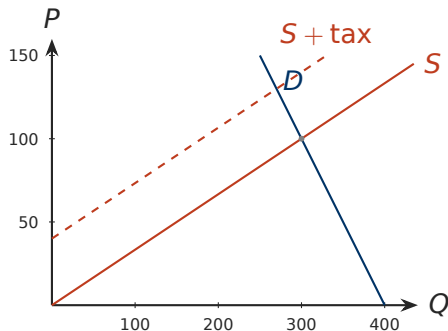
Three questions for any tax:
What happens to Q ?
What price do buyers pay, what price do sellers keep?
Does it matter who the law taxes?

A \$40 tax on sellers



Sellers owe \$40 on every unit sold

A \$40 tax on sellers

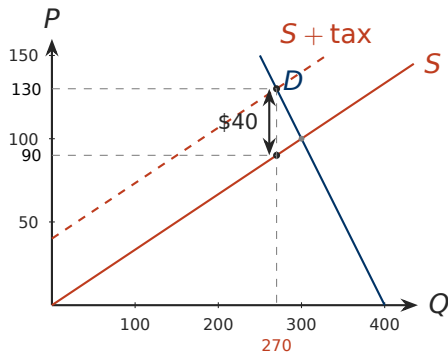


Sellers owe \$40 on every unit sold

To supply the same quantity, they need \$40 more: supply shifts **up by \$40**

$$Q_S = 3(P - 40)$$

A \$40 tax on sellers



Sellers owe \$40 on every unit sold

To supply the same quantity, they need \$40 more: supply shifts **up by \$40**

$$Q_S = 3(P - 40)$$

$$400 - P = 3P - 120 \Rightarrow P_{\text{buyers}} = \mathbf{130}$$

$$\text{Sellers keep } 130 - 40 = \mathbf{90}$$

$$Q = 400 - 130 = \mathbf{270}$$

Who paid the \$40?

	Before	After	Change
Buyers pay	\$100	\$130	+30
Sellers keep	\$100	\$90	-10
Quantity	300	270	-30

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The law taxed sellers.

Buyers carry \$30 of the \$40,
sellers \$10.

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Three results

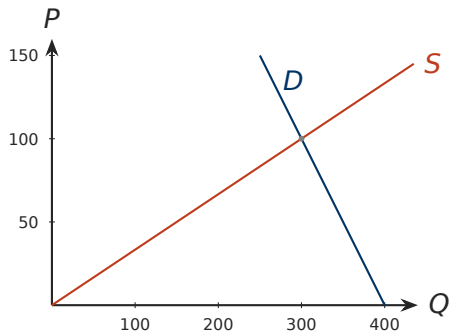
Buyers pay more, sellers keep less: **both** share the burden

The market shrinks: 30 trades that no longer happen

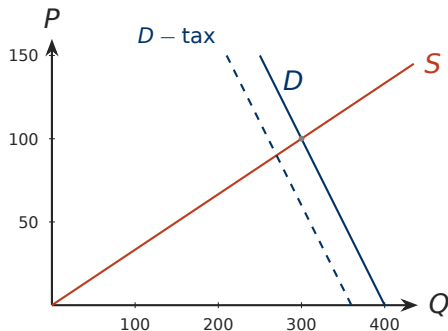
The price *wedge* between buyers and sellers equals the tax

Now tax the buyers instead

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Now tax the buyers instead

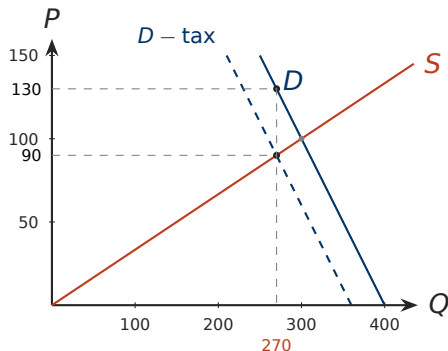


Buyers owe \$40 on every unit bought

What they will pay the seller falls by \$40: demand shifts **down by \$40**

$$Q_d = 400 - (P + 40)$$

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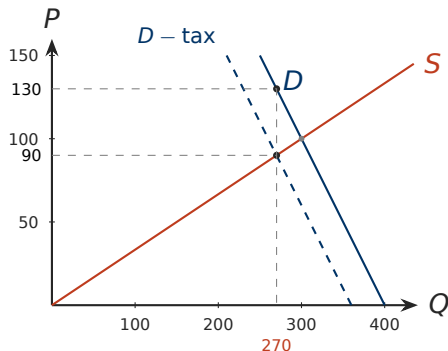
What they will pay the seller falls by \$40: demand shifts **down by \$40**

$$Q_d = 400 - (P + 40)$$

$$360 - P = 3P \Rightarrow P_{\text{sell}} = 90$$

Buyers pay $90 + 40 = 130$, $Q = 270$

Now tax the buyers instead



Buyers owe \$40 on every unit bought

What they will pay the seller falls by \$40: demand shifts **down by \$40**

$$Q_d = 400 - (P + 40)$$

$$360 - P = 3P \Rightarrow P_{\text{sell}} = \mathbf{90}$$

$$\text{Buyers pay } 90 + 40 = \mathbf{130}, \quad Q = \mathbf{270}$$

Same answer

It does not matter which side the law taxes

Can Congress split the payroll tax?

Social Security and Medicare: 15.3%
of wages

By law: 7.65% from the worker's
paycheck, 7.65% from the employer

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Question

Does the worker really pay half?

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No law can decide that. The employer's share is part of what a worker costs; it comes out of wages.

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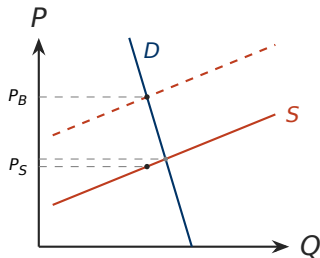
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Labor supply is fairly inelastic — most people work whatever the wage.

So workers carry most of the whole 15.3%.

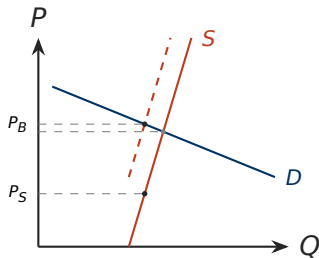
Who bears the tax: elasticity decides

Elastic S , inelastic D



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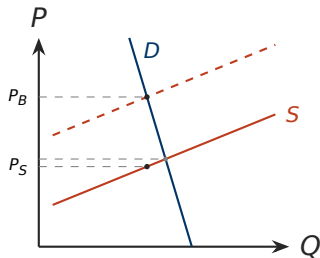
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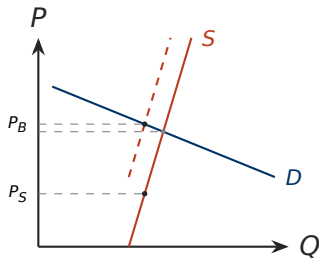
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Elastic S , inelastic D



buyers bear most

Inelastic S , elastic D



sellers bear most

The burden falls on the side that **cannot walk away** — the less elastic side.

Back to our \$40 tax

At $P = 100$, $Q = 300$:

Demand: Q changes 1 per \$1 $\Rightarrow$

$$E_d \approx \frac{1 \times 100}{300} = 1/3$$

Supply: Q changes 3 per \$1 $\Rightarrow$

$$E_s \approx \frac{3 \times 100}{300} = 1$$

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Demand is **less elastic**, so buyers carry more

Buyers \$30, sellers \$10: a 3-to-1 split, the same as $E_s : E_d = 1 : \frac{1}{3}$

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Buyers \$30, sellers \$10: a 3-to-1 split, the same as $E_s : E_d = 1 : \frac{1}{3}$

Rule of thumb

$$\text{Buyers' share} \approx \frac{E_s}{E_s + E_d}$$

$$\text{Here } \frac{1}{1 + 1/3} = \frac{3}{4} \text{ of } \$40 = \$30$$

You will not need the formula on the exam. You will need the direction: the less elastic side pays more.

Exam-style: Fenway Park

Question

Fenway seats 25,000. Demand for tickets (thousands):

$$Q_d = 100 - 1.25P.$$

- (a) Find the equilibrium price.
- (b) Boston adds a \$10 tax per ticket, paid by the buyer. Tickets sold? Fans' total price? Red Sox price? City revenue per game?
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(a) Supply is vertical at 25: $25 = 100 - 1.25P \Rightarrow P = \60 .

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(b) Still 25,000. Demand drops by \$10: Red Sox get \$50, fans pay $50 + 10 = \$60$, city $10 \times 25,000 = \$250,000$.

(c) All on the Red Sox: vertical supply passes none of it on.

Application: the 1990 luxury tax

Congress taxed yachts, private planes, furs, jewelry, expensive cars

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Burden on boatbuilders and their workers, not buyers. Most of the tax was **repealed in 1993**.

Summary

Price controls

- Ceiling = legal maximum; floor = legal minimum
- Binding only if the ceiling is *below* or the floor *above* equilibrium
- Binding ceiling $\Rightarrow$ shortage, non-price rationing; binding floor $\Rightarrow$ surplus
- Fewer units trade either way
- Rent control: shortages grow in the long run. Minimum wage: pay up for some, jobs lost for others
- Vouchers and the EITC help without fixing the price

Taxes

- A tax puts a wedge between what buyers pay and sellers keep; the market shrinks
- Tax on sellers: S up by the tax. Tax on buyers: D down by the tax.
Same outcome
- The law cannot decide who pays
- The **less elastic** side bears more of the burden

Next: review, Mon Oct 5.

Midterm 1: Wed Oct 7, Ch 1-5.